TAIPEI TECH 國立臺北科技大學 National Taipei University of Technology

# 資料結構
# Data Structure

# Lab 10

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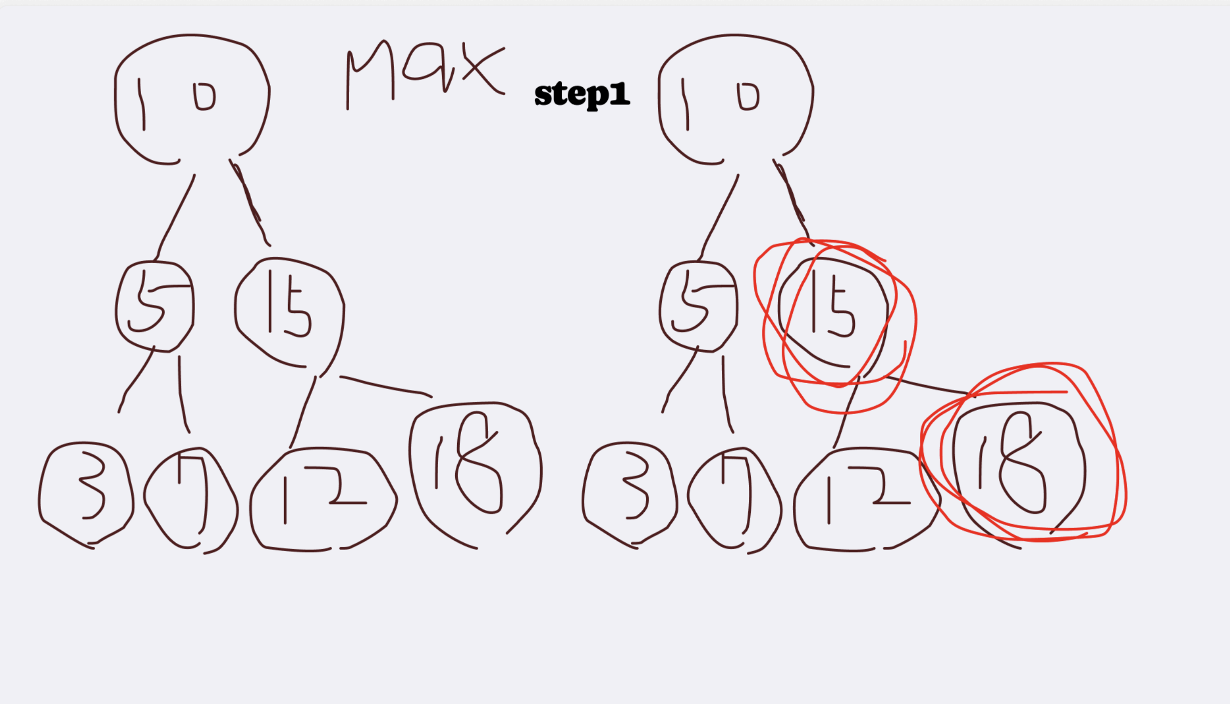
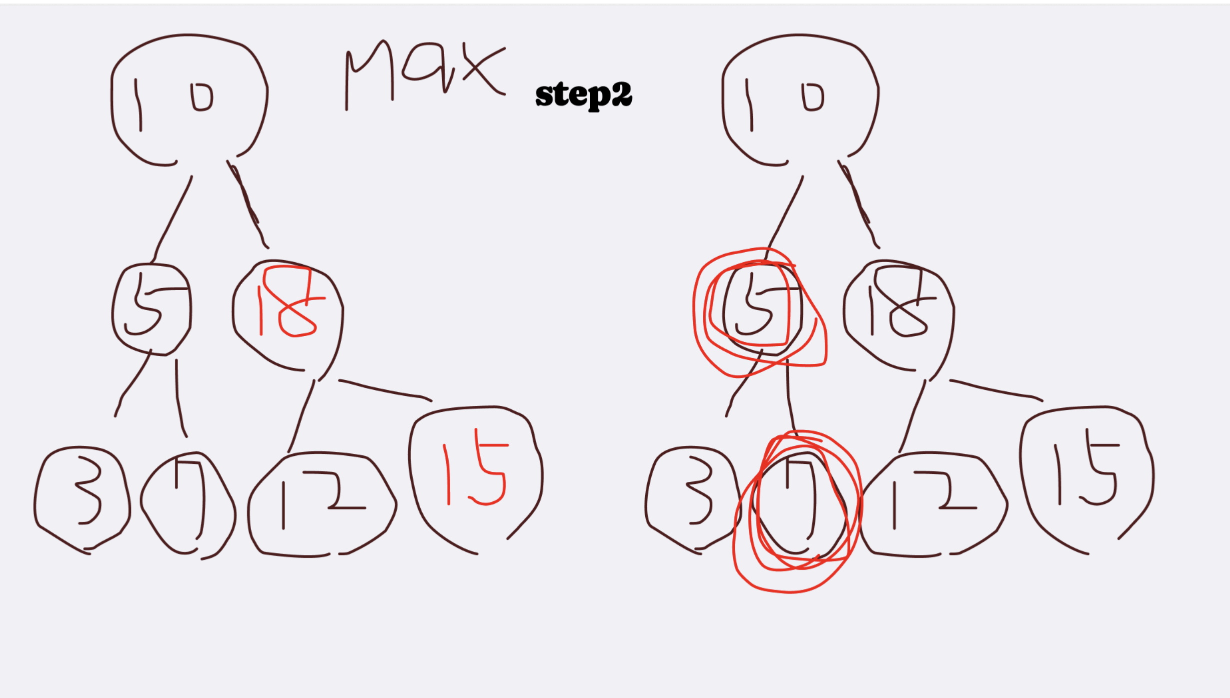
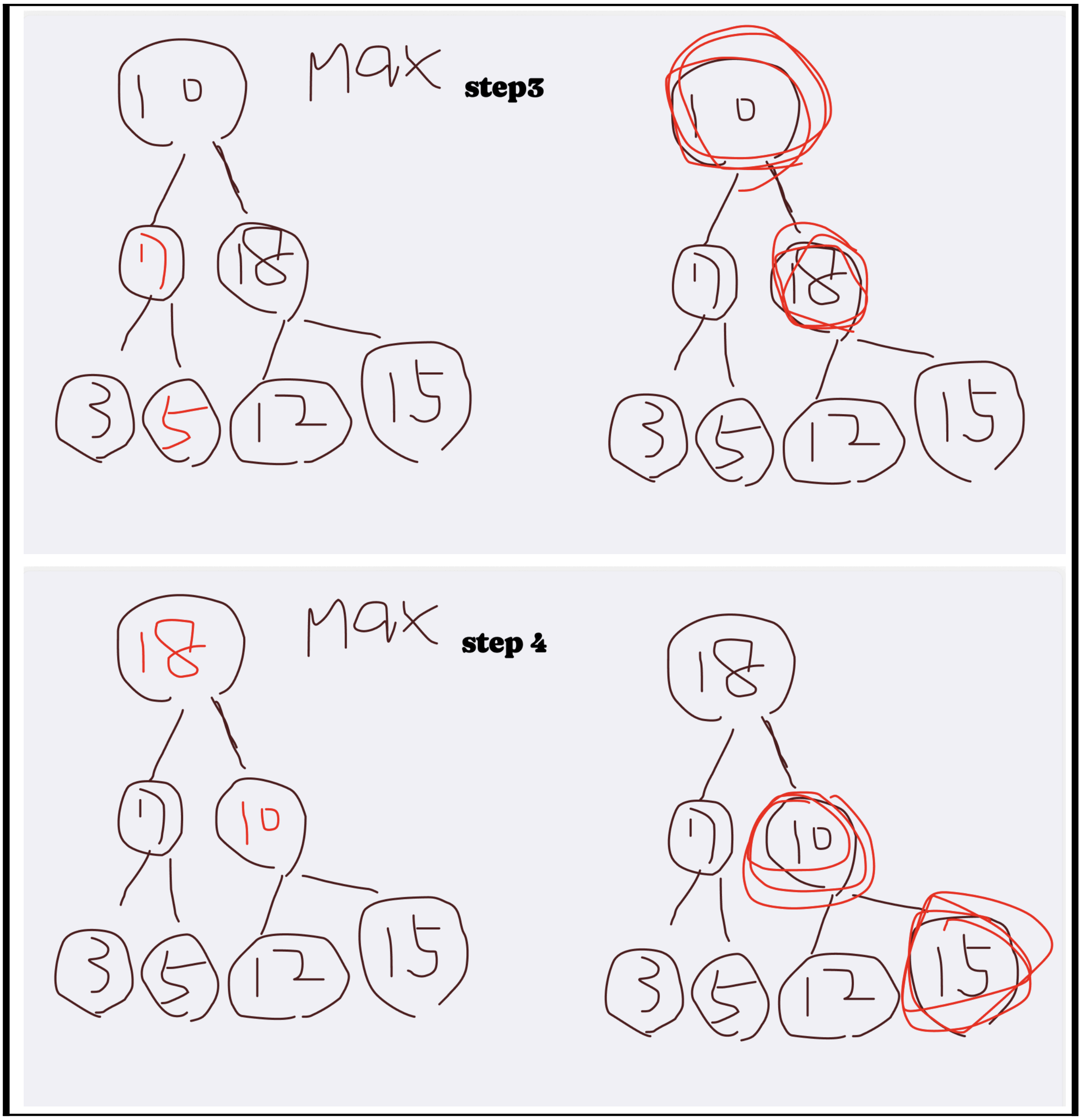
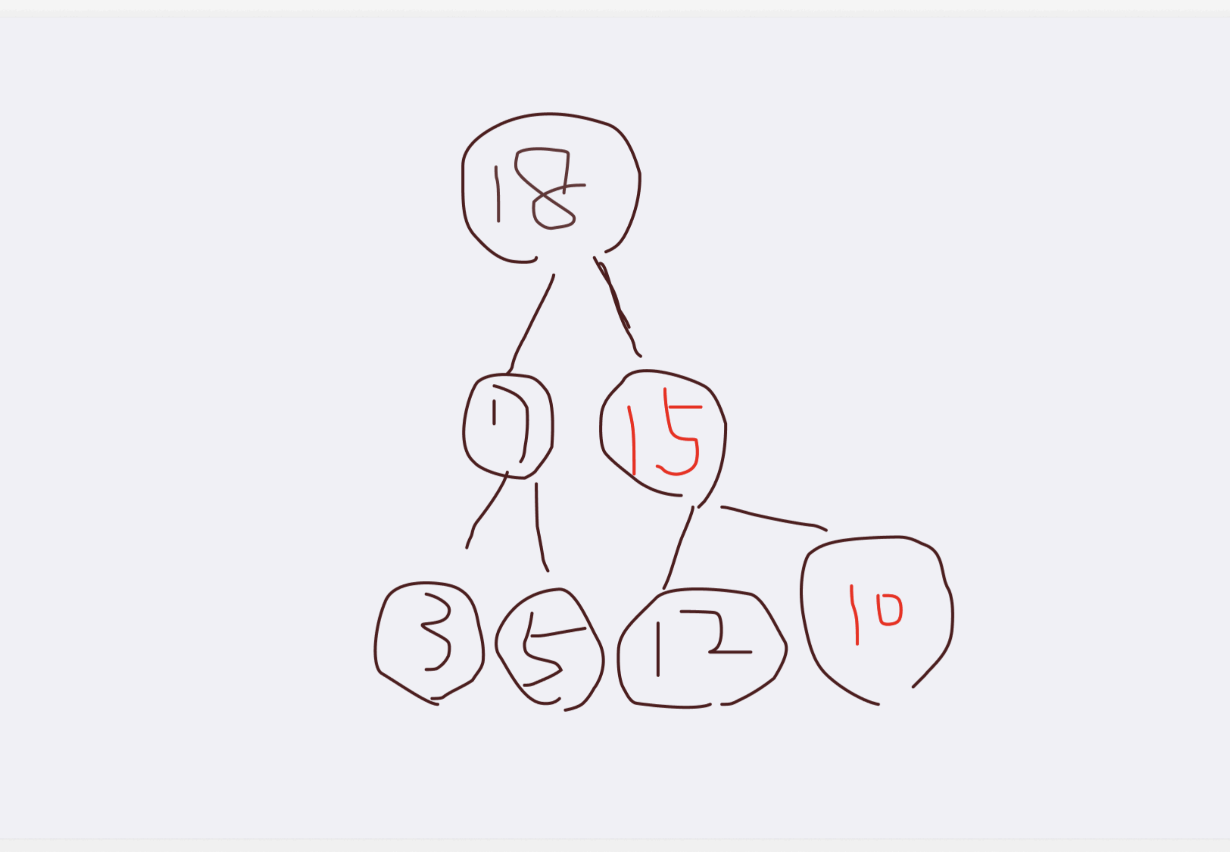
學號:112AC1015

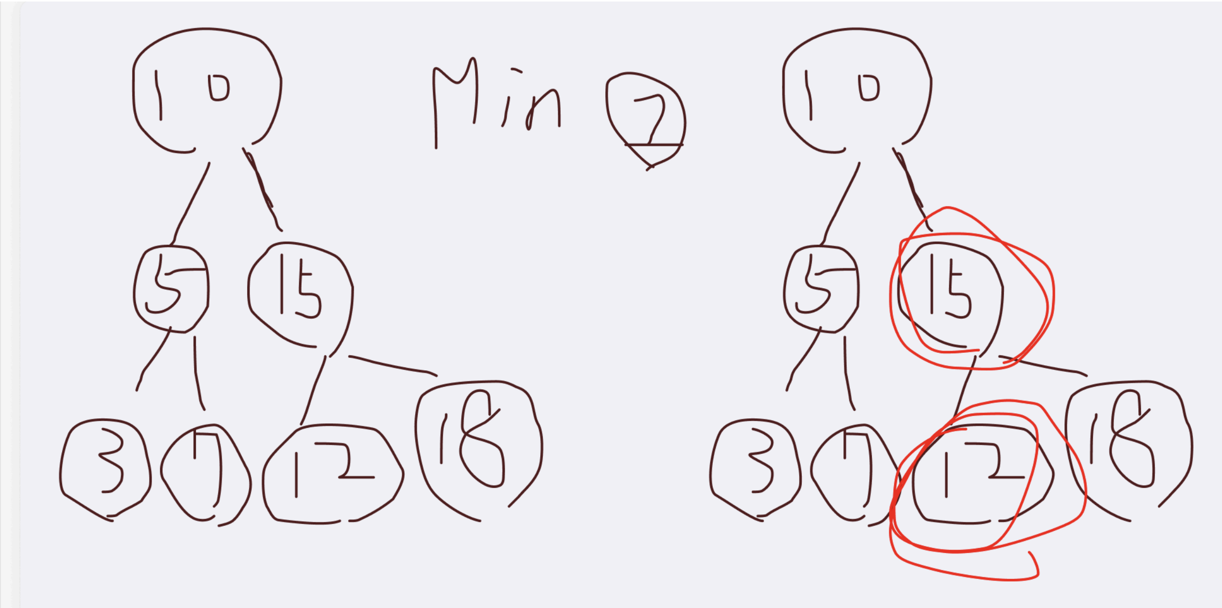
| Lab01-Ex1. 1-1 |
| --- |
| Diagrams |
| Code |

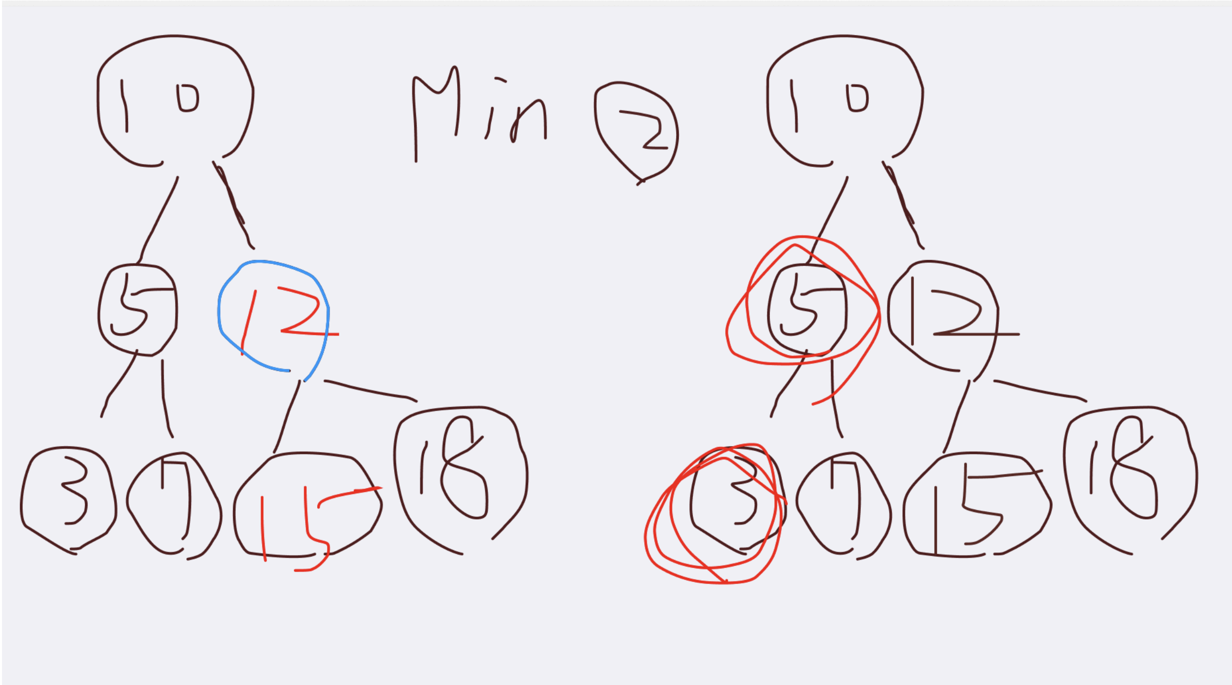


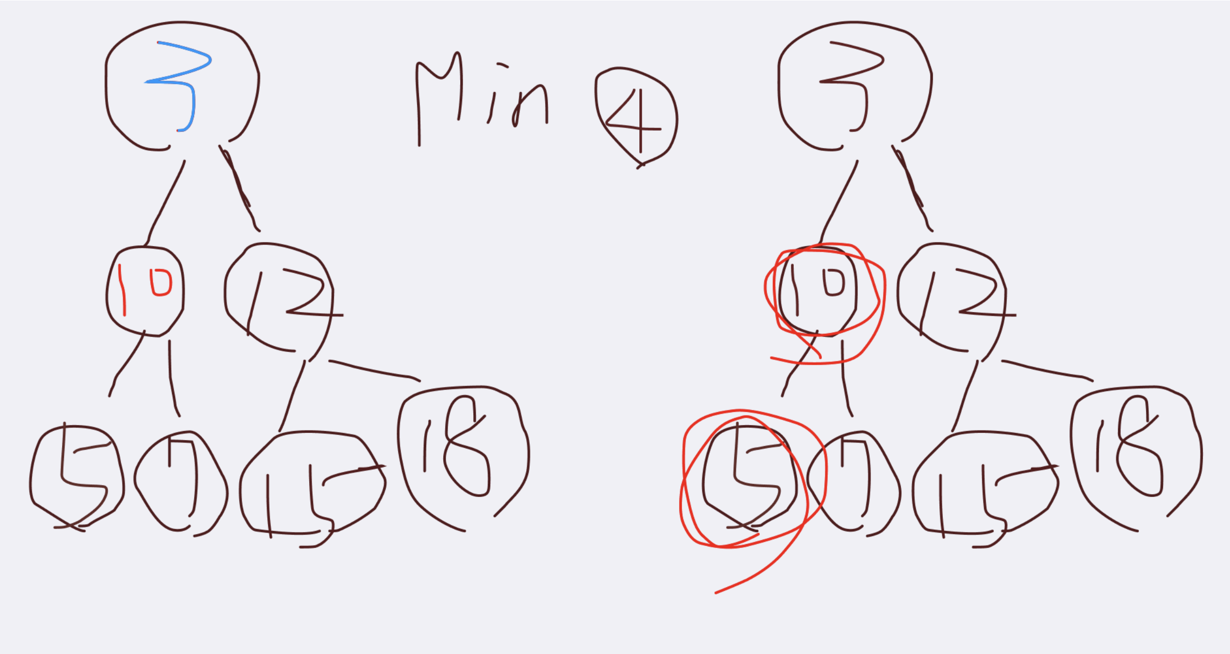

Max
step3
Max
step 4

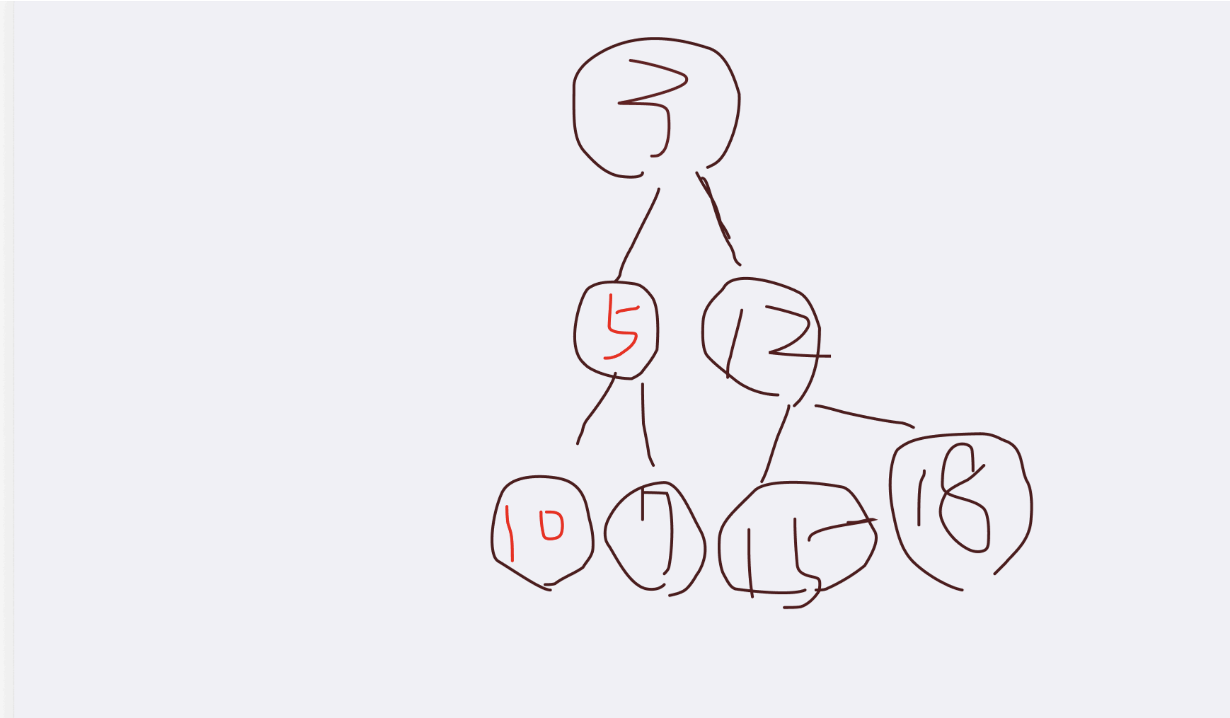

18
7
15
3
5
12
10


10
5
15
3
7
12
18
Min 7
10
5
15
3
7
12
18


Min 2
10
5
12
3
7
15
18


Min 2
10
3
12
5
7
15
18



| |
|---|
| **Lab01-Ex1. 1-2** |
| **code** |
| **Code** |

```cpp
#include <iostream>
#include <vector>
#include <fstream>
#include <sstream>
using namespace std;

// 從文件中讀取數據並存入向量
vector<int> readFromFile(const string& filename) {
    vector<int> arr;
    ifstream file(filename);

    if (!file) {
        cerr << "Error opening file: " << filename << endl;
        return arr;
    }

    string line;
    while (getline(file, line)) { // 持續讀取整行內容
        stringstream ss(line);     // 創建字符串流
        string value;
        while (getline(ss, value, ',')) { // 用逗號分隔值
            try {
                arr.push_back(stoi(value)); // 將字符串轉換為整數並存入向量
            }
            catch (exception& e) {
                cerr << "Invalid number format in file: " << value << endl;
            }
        }
    }

    file.close();
    return arr;
}

class MaxHeap {
public:
    vector<int> heap; // 儲存Max Heap的元素

    // 建立Max Heap
    void buildMaxHeap(vector<int>& arr) {  //建立Max Heap
        heap = arr;

        for (int i = (heap.size() / 2) - 1; i >= 0; i--) { // 從最後一個非葉子節點
開始向上執行Max Heap
            heapify(i);
        }
    }

    void heapify(int i) {// 堆化函式（確保以 i 為根的子樹符合Max Heap性質）
        int largest = i;           // 假設當前節點是最大的
        int left = 2 * i + 1;      // 左子節點索引
        int right = 2 * i + 2;     // 右子節點索引

        // if判斷式，檢查左子節點是否為有效範圍且比當前節點(父節點)大
        if(left < heap.size() && heap[left] > heap[largest]) {
```

```
            largest = left;//如果判斷式成立，把largest設為left
        }

        // if判斷式，檢查右子節點是否為有效範圍且比當前節點(父節點)大
        if(right < heap.size() && heap[right] > heap[largest]) {
            largest = right;//如果判斷式成立，把largest設為right
        }

        if(largest != i) {
            swap(heap[i], heap[largest]); // if判斷式，如果最大的不是父節點，交換並繼
續堆化
            heapify(largest); // 遞迴處理受影響的子樹
        }

    }

    // 顯示Heap的內容(使用BFS)
    void printHeap() {
        for (int val : heap) {// 遍歷 Max Heap中的每個元素
            cout << val << " ";// 輸出元素
        }
        cout << endl;
    }
};

int main() {
    // 從文件讀取輸入元素
    string filename = "input3.txt";//請貼上input檔案的正確路徑
    vector<int> arr = readFromFile(filename);//讀取數據

    if (arr.empty()) {// 如果數據為空
        cerr << "No valid data found in file." << endl;//輸出錯誤信息
        return -1;
    }
    cout << "Input Array: ";//輸出讀取的數據
    for (int val : arr) {
        cout << val << " ";//輸出每個元素
    }
    cout << endl;

    MaxHeap maxHeap;// 創建Max Heap對象
    maxHeap.buildMaxHeap(arr); // 建立Max Heap

    // 輸出Max Heap的內容
    cout << "Max Heap: ";
    maxHeap.printHeap();
    cout << endl;

    system("pause");
    return 0;
}
```

```
aptflin@linpipiaipupudeMacBook-Air Desktop % cd
le_code && "/Users/aptflin/Desktop/"sample_code
Input Array: 23 5 67 12 89 45 32
Max Heap: 89 23 67 12 5 45 32
```

sample_code.cpp

sample_code.cpp > MaxHeap

```cpp
class MaxHeap {
    void heapify(int i) {// 堆化函式 (確保以 i 為根的子樹符合Max Heap性質)
        int right = 2 * i + 2;    // 右子節點索引

        // if判斷式，檢查左子節點是否為有效範圍且比當前節點(父節點)大
        if(left < heap.size() && heap[left] > heap[largest]) {
            largest = left;//如果判斷式成立，把largest設為left
        }

        // if判斷式，檢查右子節點是否為有效範圍且比當前節點(父節點)大
        if(right < heap.size() && heap[right] > heap[largest]) {
            largest = right;//如果判斷式成立，把largest設為right
        }

        if(largest != i) {
            swap(heap[i], heap[largest]); // if判斷式，如果最大的不是父節點，交換並繼續堆化
            heapify(largest); // 遞迴處理受影響的子樹
        }

    }

    // 顯示Heap的內容(使用BFS)
    void printHeap() {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
cd "/Users/aptflin/Desktop/" && g++ sample_code.cpp -o sample_code && "/Users/aptflin/Desktop/"sample_c
aptflin@linpipiaipupudeMacBook-Air Desktop % cd "/Users/aptflin/Desktop/" && g++ sample_code.cpp -o samp
le_code && "/Users/aptflin/Desktop/"sample_code
Input Array: 10 5 15 3 7 12 18
Max Heap: 18 7 15 3 5 12 10
```

```
aptflin@linpipiaipupudeMacBook-Air Desktop % cd "/Users/ap
le_code && "/Users/aptflin/Desktop/"sample_code
Input Array: 10 5 15 3 7 12 18
Max Heap: 18 7 15 3 5 12 10
```

```
aptflin@linpipiaipupudeMacBook-Air Deskt
le_code && "/Users/aptflin/Desktop/"samp
Input Array: 100 80 60 40 20 10 5
Max Heap: 100 80 60 40 20 10 5
```

sample_code.cpp

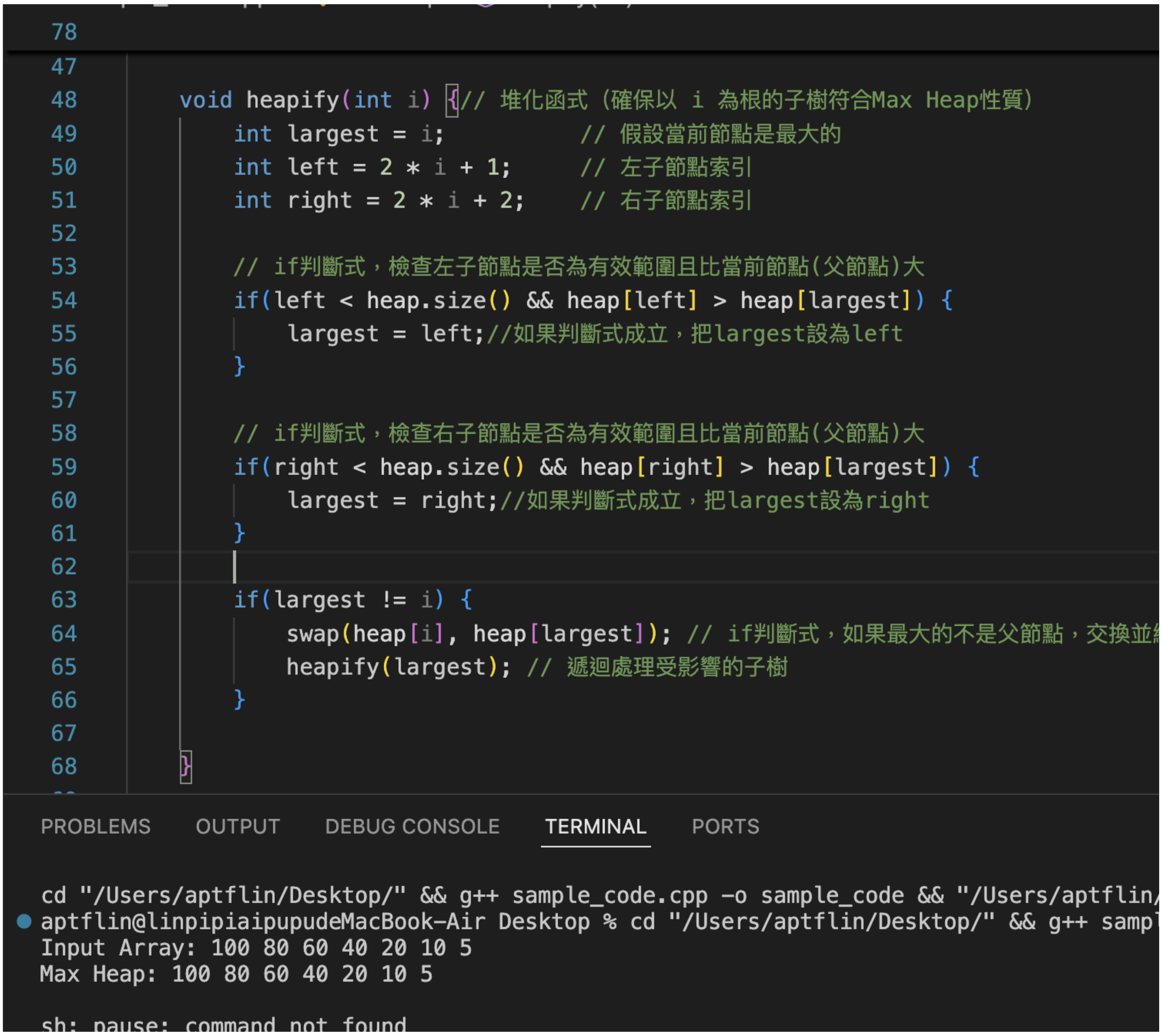
sample_code.cpp > MaxHeap > heapify(int)

```cpp
class MaxHeap {
    void heapify(int i) {// 堆化函式 (確保以 i 為根的子樹符合Max Heap性質)
        int right = 2 * i + 2;    // 右子節點索引

        // if判斷式，檢查左子節點是否為有效範圍且比當前節點(父節點)大
        if(left < heap.size() && heap[left] > heap[largest]) {
            largest = left;//如果判斷式成立，把largest設為left
        }

        // if判斷式，檢查右子節點是否為有效範圍且比當前節點(父節點)大
        if(right < heap.size() && heap[right] > heap[largest]) {
            largest = right;//如果判斷式成立，把largest設為right
        }

        if(largest != i) {
            swap(heap[i], heap[largest]); // if判斷式，如果最大的不是父節點，交換並繼續堆化
            heapify(largest); // 遞迴處理受影響的子樹
        }

    }

    // 顯示Heap的內容(使用BFS)
    void printHeap() {
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
cd "/Users/aptflin/Desktop/" && g++ sample_code.cpp -o sample_code && "/Users/aptflin/Desktop/
aptflin@linpipiaipupudeMacBook-Air Desktop % cd "/Users/aptflin/Desktop/" && g++ sample_code.c
Input Array: 23 5 67 12 89 45 32
Max Heap: 89 23 67 12 5 45 32

sh: pause: command not found
aptflin@linpipiaipupudeMacBook-Air Desktop %
```

```
sh: pause: command not found
aptflin@linpipiaipupudeMacBook-Air Deskt
le_code && "/Users/aptflin/Desktop/"samp
Input Array: 100 80 60 40 20 10 5
Max Heap: 100 80 60 40 20 10 5
```




```
#include <iostream>
#include <vector>
#include <fstream>
#include <sstream>
using namespace std;

// 從文件中讀取數據並存入向量
vector<int> readFromFile(const string& filename) {
    vector<int> arr;
    ifstream file(filename);

    if (!file) {
        cerr << "Error opening file: " << filename << endl;
        return arr;
    }

    string line;
    while (getline(file, line)) { // 持續讀取整行內容
        stringstream ss(line);     // 創建字符串流
        string value;
        while (getline(ss, value, ',')) { // 用逗號分隔值
            try {
                arr.push_back(stoi(value)); // 將字符串轉換為整數並存入向量
```

```
                }
                catch (exception& e) {
                    cerr << "Invalid number format in file: " << value << endl;
                }
            }
        }

    file.close();
    return arr;
}

class MaxHeap {
public:
    vector<int> heap; // 儲存Max Heap的元素

    // 建立Max Heap
    void buildMaxHeap(vector<int>& arr) {  //建立Max Heap
        heap = arr;

        for (int i = (heap.size() / 2) - 1; i >= 0; i--) { // 從最後一個非葉子節點
開始向上執行Max Heap
            heapify(i);
        }
    }

    void heapify(int i) {// 堆化函式（確保以 i 為根的子樹符合Max Heap性質）
        int largest = i;           // 假設當前節點是最大的
        int left = 2 * i + 1;      // 左子節點索引
        int right = 2 * i + 2;     // 右子節點索引

        // if判斷式，檢查左子節點是否為有效範圍且比當前節點(父節點)大
        if(left < heap.size() && heap[left] < heap[largest]) {
            largest = left;//如果判斷式成立，把largest設為left
        }

        // if判斷式，檢查右子節點是否為有效範圍且比當前節點(父節點)大
        if(right < heap.size() && heap[right] < heap[largest]) {
            largest = right;//如果判斷式成立，把largest設為right
        }

        if(largest != i) {
            swap(heap[i], heap[largest]); // if判斷式，如果最大的不是父節點，交換並繼
續堆化
            heapify(largest); // 遞迴處理受影響的子樹
        }

    }

    // 顯示Heap的內容(使用BFS)
    void printHeap() {
        for (int val : heap) {// 遍歷 Max Heap中的每個元素
            cout << val << " ";// 輸出元素
        }
        cout << endl;
    }
};
```

```cpp
int main() {
    // 從文件讀取輸入元素
    string filename = "input3.txt";//請貼上input檔案的正確路徑
    vector<int> arr = readFromFile(filename);//讀取數據

    if (arr.empty()) {// 如果數據為空
        cerr << "No valid data found in file." << endl;//輸出錯誤信息
        return -1;
    }
    cout << "Input Array: ";//輸出讀取的數據
    for (int val : arr) {
        cout << val << " ";//輸出每個元素
    }
    cout << endl;

    MaxHeap maxHeap;// 創建Max Heap對象
    maxHeap.buildMaxHeap(arr); // 建立Max Heap

    // 輸出Max Heap的内容
    cout << "Min Heap: ";
    maxHeap.printHeap();
    cout << endl;

    system("pause");
    return 0;
}
```

```
de
aptflin@linpipiaipupudeMacBook-Air Deskt
le_code && "/Users/aptflin/Desktop/"samp
Input Array: 10 5 15 3 7 12 18
Min Heap: 3 5 12 10 7 15 18

sh: pause: command not found
```

```cpp
int main() {
    // 從文件讀取輸入元素
    string filename = "input1.txt";//請貼上input檔案的正確路徑
    vector<int> arr = readFromFile(filename);//讀取數據

    if (arr.empty()) {// 如果數據為空
        cerr << "No valid data found in file." << endl;//輸
        return -1;
    }
    cout << "Input Array: ";//輸出讀取的數據
    for (int val : arr) {
        cout << val << " ";//輸出每個元素
    }
    cout << endl;

    MaxHeap maxHeap;// 創建Max Heap對象
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
cd "/Users/aptflin/Desktop/" && g++ sample_code.cpp -o sample_code
de
aptflin@linpipiaipupudeMacBook-Air Desktop % cd "/Users/aptflin/Des
le_code && "/Users/aptflin/Desktop/"sample_code
Input Array: 10 5 15 3 7 12 18
Min Heap: 3 5 12 10 7 15 18
```

```
sh: pause: command not found
aptflin@linpipiaipupudeMacBook-Air Des
le_code && "/Users/aptflin/Desktop/"sa
Input Array: 23 5 67 12 89 45 32
Min Heap: 5 12 32 23 89 45 67
```

```cpp
int main() {
    // 從文件讀取輸入元素
    string filename = "input2.txt";//請貼上input檔案的正確路徑
    vector<int> arr = readFromFile(filename);//讀取數據

    if (arr.empty()) {// 如果數據為空
        cerr << "No valid data found in file." << endl;//輸出錯誤信
        return -1;
    }
    cout << "Input Array: ";//輸出讀取的數據
    for (int val : arr) {
        cout << val << " ";//輸出每個元素
    }
    cout << endl;

    MaxHeap maxHeap;// 創建Max Heap對象
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
aptflin@linpipiaipupudeMacBook-Air Desktop % cd "/Users/aptflin/Desktop/"
le_code && "/Users/aptflin/Desktop/"sample_code
Input Array: 10 5 15 3 7 12 18
Min Heap: 3 5 12 10 7 15 18

sh: pause: command not found
aptflin@linpipiaipupudeMacBook-Air Desktop % cd "/Users/aptflin/Desktop/"
le_code && "/Users/aptflin/Desktop/"sample_code
Input Array: 23 5 67 12 89 45 32
Min Heap: 5 12 32 23 89 45 67

sh: pause: command not found
```

Focus folder in explorer (cm

```
sh: pause: command not found
aptflin@linpipiaipupudeMacBook-Air Deskt
le_code && "/Users/aptflin/Desktop/"samp
Input Array: 100 80 60 40 20 10 5
Min Heap: 5 20 10 40 80 100 60

sh: pause: command not found
```

```cpp
};

int main() {
    // 從文件讀取輸入元素
    string filename = "input3.txt";//請貼上input檔案的正確路徑
    vector<int> arr = readFromFile(filename);//讀取數據

    if (arr.empty()) {// 如果數據為空
        cerr << "No valid data found in file." << endl;//輸
        return -1;
    }
    cout << "Input Array: ";//輸出讀取的數據
    for (int val : arr) {
        cout << val << " ";//輸出每個元素
    }
    cout << endl;

    MaxHeap maxHeap;// 創建Max Heap對象
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
aptflin@linpipiaipupudeMacBook-Air Desktop % cd "/Users/aptflin/Des
le_code && "/Users/aptflin/Desktop/"sample_code
Min Heap: 5 12 32 23 89 45 67

sh: pause: command not found
aptflin@linpipiaipupudeMacBook-Air Desktop % cd "/Users/aptflin/Des
le_code && "/Users/aptflin/Desktop/"sample_code
Input Array: 100 80 60 40 20 10 5
Min Heap: 5 20 10 40 80 100 60

sh: pause: command not found
```